

0.1 Exam 4 Review

0.1.1 Binary and Base n

Write 1577 in binary

$$1577 = 2 \cdot 788 + 1$$

$$788 = 2 \cdot 394 + 0$$

$$394 = 2 \cdot 197 + 0$$

$$197 = 2 \cdot 98 + 1$$

$$98 = 2 \cdot 49 + 0$$

$$49 = 2 \cdot 24 + 1$$

$$24 = 2 \cdot 12 + 0$$

$$12 = 2 \cdot 6 + 0$$

$$6 = 2 \cdot 3 + 0$$

$$3 = 2 \cdot 1 + 1$$

$$1 = 2 \cdot 0 + 1$$

Writing the bottom to top left to right...

11000101001

We could check this by just doing

$$1577 = 2^{10} + 2^9 + \dots$$

Let's try representing 1577 in base 8. Remember that each remainder we get is the coefficient to the corresponding power

$$1577 = 8 \cdot 197 + 1$$

$$197 = 8 \cdot 24 + 5$$

$$24 = 8 \cdot 3 + 0$$

$$3 = 8 \cdot 0 + 3$$

Rewriting this, we get

$$\underbrace{3}_{8^3} \underbrace{0}_{8^3} \underbrace{5}_{8^2} \underbrace{1}_{8^0}$$

So

$$1577 = 3 \cdot 8^3 + 5 \cdot 8^1 + 1 \cdot 8^0$$

Now let's do it with base 16

$$1577 = 16 \cdot 98 + 9$$

$$98 = 16 \cdot 6 + 2$$

$$6 = 16 \cdot 0 + 6$$

$$\boxed{1577 = 629_{16}}$$

$$1577 = 6 \cdot 16^2 + 2 \cdot 16^1 + 9 \cdot 16^0$$

For our last example, write the $AB0CD_{16}$ in base 10 (aka normal numbers)

Recall that

$$A = 10$$

$$B = 11$$

$$\vdots$$

$AB0CD$ is just 10 11 0 12 13

Since there's 5 "numbers" that means we start with $16^{5-1} = 16^4$

$$\begin{aligned} AB0CD_{16} &= 10 \cdot 16^4 + 11 \cdot 16^3 + 0 \cdot 16^2 + 12 \cdot 16^1 + 13 \cdot 16^0 \\ &= \boxed{700621} \quad \text{(just plug into calc)} \end{aligned}$$

0.1.2 RSA and Encryption

FLT Encrytion

For Encryption using FLT, basically

We have:

M : Message
 C : Encrypted Message
 e : Encryption Key
 d : Decryption Key
 p : Prime mod

To **encrypt** a message M to C

$$C \equiv M^e \pmod{p}$$

To **decrypt** C back into M , we must first find d

$$ed \equiv 1 \pmod{p-1}$$

Just find e^{-1} using Euclidean and plug it into

$$M \equiv C^d \pmod{p}$$

FLT / RSA Encryption

We have:

M : Message
 C : Encrypted Message
 e : Encryption Key
 d : Decryption Key
 p : Prime mod 1
 q : Prime mod 2
 N : Product of $p \cdot q$

To **encrypt** a message M to C

$$C \equiv M^e \pmod{N}$$

To **decrypt** C back into M , we must first find d

$$ed \equiv 1 \pmod{(p-1)(q-1)}$$

Just find e^{-1} using Euclidean and plug it into

$$M \equiv C^d \pmod{N}$$

0.1.3 Proof By Induction

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Proof

BASE CASE:

$$\begin{aligned} \sum_{k=1}^1 k^3 &= 1^3 = 1 \\ &= \left[\frac{n(n+1)}{2} \right]^2 = \left[\frac{1(1+1)}{2} \right]^2 = 1 \end{aligned}$$

So the case $n = 1$ is true

INDUCTIVE CASE:

Assume that $\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$

We then prove $\sum_{k=1}^{n+1} k^3 = \left[\frac{n+1(n+2)}{2} \right]^2$

Let's do this by making the more complicated part (the proving part) equal to the simpler one

$$\begin{aligned}
\sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^n k^3 + (n+1)^3 = \sum_{k=1}^{n+1} k^3 = \left[\frac{n(n+2)}{2} \right]^2 + (n+1)^3 \\
&= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\
&= (n+1)^2 \cdot \left[\frac{n^2}{4} + (n+1) \right] \\
&= (n+1)^2 \cdot \left[\frac{n^2 + 4n + 4}{4} \right] \\
&= (n+1)^2 \cdot \frac{(n+2)^2}{4} \\
&= \frac{(n+1)^2(n+2)^2}{4} \\
&= \boxed{\left[\frac{(n+1)(n+2)}{2} \right]^2}
\end{aligned}$$

0.1.4 Multiples of n up to k

Theorem 1 (Number of Multiples of n up to k)

To find the number of positive integers divisible by n up to k is given by

$$\left\lfloor \frac{k}{n} \right\rfloor$$

This DOES NOT include 0, so for all whole numbers it would be

$$\left\lfloor \frac{k}{n} \right\rfloor + 1$$

0.1.5 Pigeonhole Principle

Definition 1 (Pigeonhole Principle)

If there are n boxes and more than n objects placed in these boxes, at least one box contains more than one object.

At least one box will contain

$$\left\lceil \frac{k}{n} \right\rceil$$

objects

0.1.6 Permutaions and Combinations

Permutations

Definition 2 (Permutation)

A **Permutation** is one specific way a group of items can be arranged where order matters. In general, if we have n objects and we want to arrange k of them, then the number of ways is given by

$${}_nP_k = \frac{n!}{(n-k)!}$$

Combinations

Definition 3 (Combination)

A **Combinaton** is just an arrangement where order does not matter.

The number of combinations of k items taken from n items is

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

You would say "n choose k"

Think of this as "how many ways can you arrange k things in n slots"

0.1.7 Binomial Theorem

Theorem 2 (Binomial Theorem)

$$\text{For } n = 0, 1, 2, \dots, (a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

0.1.8 Sum of Combinations

Theorem 3 (Sum of Combinations)

$$\text{For } n = 0, 1, 2, \dots, \sum_{k=0}^n \binom{n}{k} = 2^n$$

0.1.9 Arrangements of Letters

For a word with k letters, where each letter has n_i appearances, the numbers of arrangements is given by

$$\frac{k!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

You basically just do

$$\binom{k}{n_1} \cdot \binom{k-n_1}{n_2} \cdot \binom{k-n_1-n_2}{n_3} \cdot \dots$$

For a word with all unique letters, it's just

$$k!$$

0.1.10 Stars and Bars

When you have n slots and k options for each slot, the number of possible combinations is

$$\frac{(n+k-1)!}{n! \cdot (k-1)!} = \binom{n+k-1}{k-1}$$

Another way of thinking of this is stars and bars

Let

s : Stars, num slots

b : Bars, options - 1

Then the combinations is

$$\frac{(s+b)!}{s! \cdot b!} = \binom{s+b}{b}$$

0.1.11 Binomial Probability

Given a limited event (ie 2 outcomes) with probability p ,

$$p(k \text{ successes in } n \text{ trials}) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

0.1.12 Expected Value (EV)

$$EV = \sum (\text{probability})(\text{payoff})$$